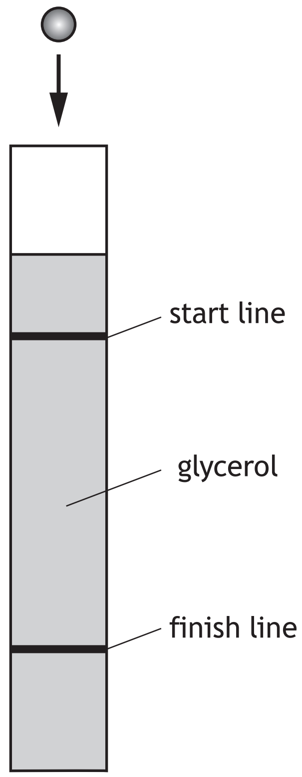
15. A student carries out an experiment to measure the terminal velocity of ball bearings with different diameters falling through glycerol.

Each ball bearing is dropped into a long tube filled with glycerol.



(a) Explain in terms of the forces acting on the ball bearing, why it reaches its terminal velocity. **2**

[Turn over

15. (continued) MARKS

(b) The student measures the diameter d of each ball bearing and records the corresponding terminal velocity v_t.

The results are shown in the table.

d (m)	d^2 (m^2)	v_t ($m\,s^{-1}$)
$3{\cdot}15 \times 10^{-3}$	$0{\cdot}99 \times 10^{-5}$	0·05
$4{\cdot}77 \times 10^{-3}$	$2{\cdot}28 \times 10^{-5}$	0·10
$6{\cdot}34 \times 10^{-3}$	$4{\cdot}02 \times 10^{-5}$	0·18
$9{\cdot}52 \times 10^{-3}$	$9{\cdot}06 \times 10^{-5}$	0·32
$12{\cdot}65 \times 10^{-3}$	$16{\cdot}00 \times 10^{-5}$	0·52

(i) Using the square-ruled paper on *page 42*, draw a graph of v_t against d^2. **3**

(The table of results is also shown on *page 43*, opposite the square-ruled paper.)

(ii) The student suspects that the results show that there is a systematic uncertainty in the measurements.

Suggest a reason why the student has come to this conclusion. **1**

(iii) Calculate the gradient of your graph. **2**

Space for working and answer

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15. (b) (continued)

(iv) The terminal velocity v_t of each ball bearing is given by

$$v_t = \frac{375g}{\eta} \times d^2$$

where η is the viscosity of the glycerol in pascal seconds (Pa s)

d is the diameter of the ball bearing in m

g is gravitational field strength on Earth in N kg^{-1}.

Use the gradient of your graph to determine the viscosity of the glycerol. **2**

Space for working and answer

[END OF QUESTION PAPER]

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